

# Amrut Prajapati

AI/ML Engineer · Ahmedabad, India

Computer vision and agentic AI, engineered for reliability.

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Portfolio [amrut-prajapati.github.io](https://amrut-prajapati.github.io)

GitHub [github.com/Amrut-Prajapati](https://github.com/Amrut-Prajapati)

LinkedIn [linkedin.com/in/amrut-prajapati](https://linkedin.com/in/amrut-prajapati)

Based in Ahmedabad, India · remote-ready

5,000+

automated tests

on a real-time video analytics platform

< 20 ms

per frame

real-time 3D mapping

₹2,00,000

scholarship

\_VOIS Innovation Marathon 2.0 winner

7B

vision-language model

fine-tuned and run on a single GPU

## PROFILE

AI/ML engineer who builds computer-vision, vision-language and agentic systems that keep working after the demo: held-out testing, thousands of automated tests, multi-hour soak runs, and models engineered for modest hardware. Winner of the ₹2,00,000 \_VOIS Innovation Marathon 2.0 with ReUpyog, a confidential flagship project. Final-year B.Tech CSE, GLS University.

## SELECTED WORK

### ReUpyog

Flagship project · A confidential AI project

- Winner of the ₹2,00,000 \_VOIS Innovation Marathon 2.0.
- SSIP seed grant, Government of Gujarat.
- Capstone Project I and II at GLS University, both graded O+.
- Technical details are confidential and available to serious enquiries on request.

### CONFIDENTIAL BUILDS

#### Real-time video analytics platform

GPU-accelerated, with an operator console; multi-hour soak tests and 5,000+ automated tests.

#### Real-time 3D perception & mapping system

Segmentation network trained from scratch; GPU mapping under 20 ms per frame; 3D web console.

#### Agentic vision-language assistant

Fine-tuned 7B model on one GPU; routes questions to vision tools; evidence and a confidence score with every answer.

#### Technology-intelligence & forecasting platform · SIH 2025 finalist

Automates collection, analysis and long-range forecasting of strategic-technology trends; offline GPU AI engine; executive dashboard.

## CAPABILITIES

**Computer vision & 3D.** YOLO family, RT-DETR, SAM2, OpenCV, segmentation, multi-object tracking, OCR, LiDAR and point clouds

**Vision-language & LLMs.** Fine-tuning (QLoRA / PEFT), tool-calling agents, RAG, local LLMs, prompt engineering, Whisper

**Machine learning.** PyTorch, TensorFlow, deep learning, model evaluation

**Deployment & backend.** FastAPI, PostgreSQL, Redis, Docker, ONNX, TensorRT, CUDA, ROS 2, n8n

**Frontend & testing.** React, Next.js, TypeScript, Three.js · pytest, Hypothesis, Playwright

**Languages.** Python, Java, C++, JavaScript, SQL · English, Hindi, Gujarati

## RECOGNITION

**Winner, \_VOIS Innovation Marathon 2.0** · Feb 2026

**SSIP seed grant, Government of Gujarat** · Mar 2026

**Finalist, Smart India Hackathon 2025** · 2025

## CREDENTIALS

### OCI Generative AI Professional

Oracle, Oct 2025

### OCI AI Foundations Associate

Oracle, Oct 2025

### OCI Data Science Professional

Oracle, Oct 2025

### Core Java Programming

GLS University MOOC, grade O+

## EDUCATION

**B.Tech, Computer Science & Engineering** · 2023 – 2027 (expected)

GLS University, Faculty of Engineering & Technology, Ahmedabad

CGPA 7.95 / 10 · SGPA 8.4+ in each of the last three semesters